ELITE IGCSE ACADEMY

Student Past Paper Solutions

Jan 2020 P2H Worked Solutions

Jan 2020 | P2H

33 questions ▪ question image followed by worked solution

PREPARED BY

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STUDENT EDITION

Worked answers included

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PAST PAPER QUESTION

Q#: 1 **Marks: 3**

TOPIC: **Algebraic Roots & Indices**

Jan 2020 P2H - Jan 2020 | P2H

1 (a) Simplify $\dfrac{x^9}{x^2}$

....

(1)

(b) Write $\dfrac{7^8 \times 7^4}{7^3}$ as a single power of 7

....

(2)

(Total for Question 1 is 3 marks)

PAST PAPER SOLUTION

Q#: 1 Marks: 3

TOPIC: **Algebraic Roots & Indices**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION

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Topic check. Corrected to Algebraic Roots & Indices.

Solution

(a)

$$\frac{x^9}{x^2} = x^{9-2} = x^7$$

(b)

$$\frac{7^8 \times 7^4}{7^3} = 7^{8+4-3} = 7^9$$

FINAL ANSWER

(a) x^7, (b) 7^9.

PAST PAPER QUESTION

Q#: 2 Marks: 2

Topic: **Standard & Compound Units**

Jan 2020 P2H - Jan 2020 | P2H

2 Change 32.4 m^3 into cm^3

.... cm^3

(Total for Question 2 is 2 marks)

PAST PAPER SOLUTION

Q#: 2 Marks: 2

TOPIC: **Standard & Compound Units**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION

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Topic check. Correct. This is converting cubic units.

Solution

$$1 \text{ m} = 100 \text{ cm}$$

So

$$1 \text{ m}^3 = 100^3 \text{ cm}^3 = 1,000,000 \text{ cm}^3$$

$$32.4 \text{ m}^3 = 32.4 \times 1,000,000$$

$$= 32,400,000 \text{ cm}^3$$

FINAL ANSWER

$32,400,000 \text{ cm}^3$.

PAST PAPER QUESTION

Q#: 3 Marks: 3

TOPIC: **Fractions**

Jan 2020 P2H - Jan 2020 | P2H

3 Show that $4\frac{2}{3} + 3\frac{4}{5} = 8\frac{7}{15}$

(Total for Question 3 is 3 marks)

PAST PAPER SOLUTION Q#: 3 Marks: 3

TOPIC: **Fractions**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION

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Topic check. Fractions. The tag is correct.

Method

Convert to improper fractions:

$$4\frac{2}{3} = \frac{14}{3}$$

$$3\frac{4}{5} = \frac{19}{5}$$

Use a common denominator:

$$\frac{14}{3} + \frac{19}{5} = \frac{70}{15} + \frac{57}{15}$$

$$= \frac{127}{15} = 8\frac{7}{15}$$

FINAL ANSWER

$8\frac{7}{15}$

PAST PAPER QUESTION

Q#: 4 Marks: 4

TOPIC: **Angles in Polygons & Parallel Lines**

Jan 2020 P2H - Jan 2020 | P2H

4 The diagram shows a triangle.

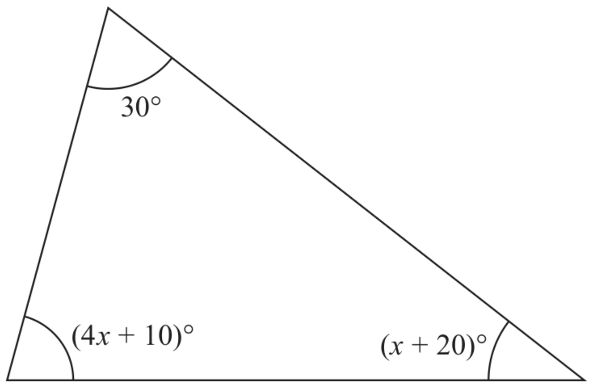

Diagram **NOT** accurately drawn

Work out the value of x.

$x =$ ....

(Total for Question 4 is 4 marks)

PAST PAPER SOLUTION

Q#: 4 Marks: 4

TOPIC: **Angles in Polygons & Parallel Lines**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION

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Topic check. Corrected to Angles in Polygons & Parallel Lines. This is a triangle angle-sum question.

Solution

Angles in a triangle add to 180°:

$$30 + (4x + 10) + (x + 20) = 180$$

$$5x + 60 = 180$$

$$5x = 120$$

$$x = 24$$

FINAL ANSWER

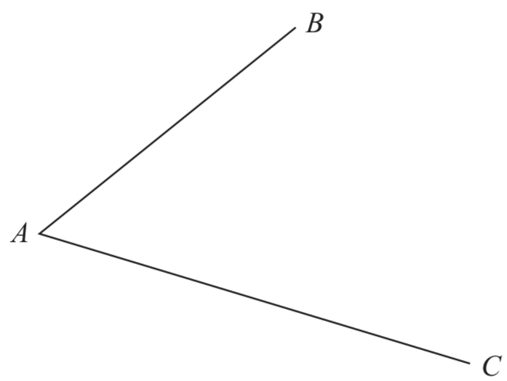
$x = 24$.

PAST PAPER QUESTION Q#: 5 Marks: 2

TOPIC: **Bearings, Scale Drawing & Constructions**

Jan 2020 P2H - Jan 2020 | P2H

5 Use ruler and compasses to construct the bisector of angle BAC.
You must show all your construction lines.



(Total for Question 5 is 2 marks)

PAST PAPER SOLUTION

Q#: 5 Marks: 2

TOPIC: **Bearings, Scale Drawing & Constructions**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION

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Topic check. Bearings, Scale Drawing & Constructions. The tag is correct.

Construction answer

To construct the bisector of angle BAC:

1. Put the compass point at A and draw an arc that cuts AB and AC.
2. From the two points where the arc cuts the arms, draw equal arcs inside the angle.
3. Draw a straight line from A through the intersection of those two arcs.

FINAL ANSWER

The ray from A through the arc intersection is the angle bisector of BAC.

PAST PAPER QUESTION

Q#: 6 Marks: 4

TOPIC: **Probability Toolkit**

Jan 2020 P2H - Jan 2020 | P2H

6 A bag contains only red beads, blue beads, green beads and yellow beads.

The table gives the probabilities that, when a bead is taken at random from the bag, the bead will be blue or the bead will be yellow.

Colour	**red**	**blue**	**green**	**yellow**
Probability		0.24		0.31

The probability that the bead will be green is twice the probability that the bead will be red.

Sofia takes at random a bead from the bag.
She writes down the colour of the bead and puts the bead back into the bag.

She does this 180 times.

Work out an estimate for the number of times she takes a red bead from the bag.

....

(Total for Question 6 is 4 marks)

PAST PAPER SOLUTION

Q#: 6 Marks: 4

TOPIC: **Probability Toolkit**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION

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Topic check. Probability Toolkit. The tag is correct.

Solution

Let the probability of red be r. Then the probability of green is $2r$.

$$r + 0.24 + 2r + 0.31 = 1$$

$$3r = 0.45$$

$$r = 0.15$$

Expected number of red counters:

$$180 \times 0.15 = 27$$

FINAL ANSWER

27.

PAST PAPER QUESTION

Q#: 7 Marks: 5

TOPIC: **Solving Quadratic Equations**

Jan 2020 P2H - Jan 2020 | P2H

7 (a) Solve the inequality $2x + 7 > 4$

....

(2)

(b) Solve $x^2 - 3x - 40 = 0$

Show clear algebraic working.

....

(3)

(Total for Question 7 is 5 marks)

PAST PAPER SOLUTION

Q#: 7 Marks: 5

TOPIC: **Solving Quadratic Equations**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION

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Topic check. Corrected to solving quadratic equations. The quadratic equation carries the larger part of the question.

Solution

(a)

$$2x + 7 > 4$$

$$2x > -3$$

$$x > -\frac{3}{2}$$

(b)

$$x^2 - 3x - 40 = 0$$

Factorise:

$$(x - 8)(x + 5) = 0$$

So

$$x = 8 \quad \text{or} \quad x = -5$$

FINAL ANSWER

$x > -\frac{3}{2}$, and $x = 8$ or $x = -5$.

PAST PAPER QUESTION

Q#: 8 Marks: 7

TOPIC: **Percentages**

Jan 2020 P2H - Jan 2020 | P2H

8 The table shows the cost, in euros, of Brigitte's car insurance in each of the years 2016, 2017 and 2018

Year	**2016**	**2017**	**2018**
Cost of insurance (euros)	500	545	592

Brigitte says,

"The percentage increase in the cost of my car insurance from 2017 to 2018 is more than the percentage increase in the cost of my car insurance from 2016 to 2017"

(a) Is Brigitte correct?
You must show how you get your answer.

(4)

Henri wants to insure his car.

He gets a discount of 15% off the normal price.
Henri pays 952 euros for his car insurance after the discount.

(b) Work out the discount that Henri gets.

.... euros

(3)

(Total for Question 8 is 7 marks)

PAST PAPER SOLUTION

Q#: 8 Marks: 7

TOPIC: **Percentages**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION

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Topic check. Corrected to Percentages. The question compares percentage changes and a discount.

Method

From 2016 to 2017, the increase is

$$545 - 500 = 45$$

$$\frac{45}{500} \times 100 = 9\%$$

From 2017 to 2018, the increase is

$$592 - 545 = 47$$

$$\frac{47}{545} \times 100 \approx 8.62\%$$

So Brigitte is not correct because the second percentage increase is smaller.
For Henri, 952 is 85% of the original price because there is a 15% discount.

$$\text{original price} = 952 \div 0.85 = 1120$$

The discount is

$$1120 - 952 = 168$$

FINAL ANSWER

Brigitte is not correct; the discount is 168 euros.

PAST PAPER QUESTION Q#: 9 Marks: 2

Topic: **Standard & Compound Units**

Jan 2020 P2H - Jan 2020 | P2H

9 The density of gold is 19.3 g/cm^3
A gold bar has volume 150 cm^3

Work out the mass of the gold bar.

.... g

(Total for Question 9 is 2 marks)

PAST PAPER SOLUTION

Q#: 9 Marks: 2

TOPIC: **Standard & Compound Units**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION

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Topic check. Correct. This is a density question.

Solution

$$\text{mass} = \text{density} \times \text{volume}$$

$$\text{mass} = 19.3 \times 150$$

$$\text{mass} = 2895$$

FINAL ANSWER

2895 g.

PAST PAPER QUESTION Q#: 10 Marks: 3

TOPIC: **Standard & Compound Units**

Jan 2020 P2H - Jan 2020 | P2H

10 Change a speed of 50 metres per second to a speed in kilometres per hour.

.... kilometres per hour

(Total for Question 10 is 3 marks)

PAST PAPER SOLUTION

Q#: 10 Marks: 3

TOPIC: **Standard & Compound Units**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION

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Topic check. Correct. This is converting metres per second to kilometres per hour.

Solution

To convert m/s to km/h, multiply by 3.6.

$$50 \times 3.6 = 180$$

FINAL ANSWER

180 km/h.

PAST PAPER QUESTION

Q#: 11 **Marks: 5**

Topic: **Area & Perimeter**

Jan 2020 P2H - Jan 2020 | P2H

11 The diagram shows a shaded shape *ABCD* made from a semicircle *ABC* and a right-angled triangle *ACD*.

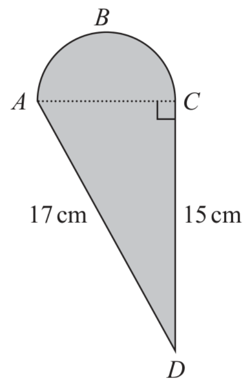

Diagram **NOT** accurately drawn

AC is the diameter of the semicircle *ABC*.

Work out the perimeter of the shaded shape.
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 11 is 5 marks)

PAST PAPER SOLUTION

Q#: 11 Marks: 5

TOPIC: **Area & Perimeter**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION

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Topic check. Area & Perimeter. The tag is correct.

Solution

Triangle ACD is right-angled, with

$$AD = 17 \text{ cm}, \qquad CD = 15 \text{ cm}$$

Use Pythagoras to find AC:

$$AC^2 = 17^2 - 15^2 = 289 - 225 = 64$$

$$AC = 8 \text{ cm}$$

So the semicircle has radius

$$r = 4 \text{ cm}$$

The arc length of the semicircle is

$$\pi r = 4\pi$$

The perimeter of the shaded shape is

$$AD + CD + \text{semicircle arc} = 17 + 15 + 4\pi = 44.566\ldots$$

FINAL ANSWER

44.6 cm.

PAST PAPER QUESTION

Topic: **Exchange Rates & Best Buys**

Jan 2020 P2H - Jan 2020 | P2H

Q#: 12 Marks: 4

12 Astrid wants to buy some oil.
She can buy the oil from either Dane Oil or Arctic Oil.

Here is information about the price that each company will charge Astrid.

Dane Oil	**Arctic Oil**
(4.2×10^5) litres for 2 500 000 Krone	(8.6×10^5) litres for 770 000 Dollars

Astrid wants to get the better value for money for the oil.

1 Dollar = 6.57 Krone

From which company should she buy her oil, Dane Oil or Arctic Oil?
You must show your working.

(Total for Question 12 is 4 marks)

PAST PAPER SOLUTION

Q#: 12 Marks: 4

TOPIC: **Exchange Rates & Best Buys**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION

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Topic check. Corrected to Exchange Rates & Best Buys. The question is mainly about comparing value for money using an exchange rate; standard form is secondary.

Method

Dane Oil costs 2500000 Krone for

$$4.2 \times 10^5 = 420000$$

litres.

$$\text{Dane cost per litre} = \frac{2500000}{420000} = 5.952\ldots$$

So Dane Oil costs about 5.95 Krone per litre.
Arctic Oil costs 770000 dollars. Convert this to Krone:

$$770000 \times 6.57 = 5058900$$

Arctic Oil gives

$$8.6 \times 10^5 = 860000$$

litres.

$$\text{Arctic cost per litre} = \frac{5058900}{860000} = 5.882\ldots$$

Arctic Oil has the lower cost per litre.

FINAL ANSWER
Arctic Oil

PAST PAPER QUESTION Q#: 13 Marks: 4

Topic: **Circle Theorems**

Jan 2020 P2H - Jan 2020 | P2H

13

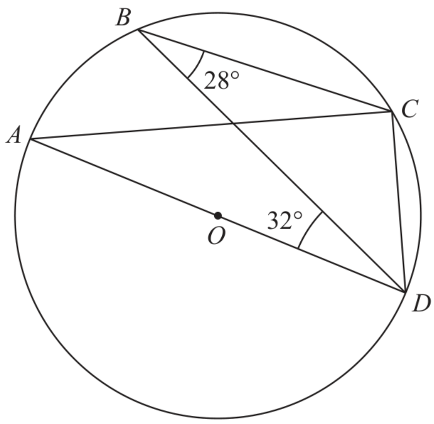

Diagram **NOT** accurately drawn

A, *B*, *C* and *D* are points on a circle, centre *O*.
AOD is a diameter of the circle.

Angle $CBD = 28°$
Angle $BDA = 32°$

Find the size of angle *BDC*.
Give a reason for each stage of your working.

.... °

(Total for Question 13 is 4 marks)

PAST PAPER SOLUTION

Q#: 13 Marks: 4

TOPIC: **Circle Theorems**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION

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Topic check. Circle Theorems. The tag is correct.

Solution

Since AD is a diameter,

$$\angle ABD = 90^\circ$$

Given

$$\angle CBD = 28^\circ$$

So

$$\angle ABC = 90^\circ - 28^\circ = 62^\circ$$

Opposite angles in a cyclic quadrilateral add to 180°:

$$\angle ADC = 180^\circ - 62^\circ = 118^\circ$$

At D,

$$\angle ADC = \angle BDA + \angle BDC$$

$$118^\circ = 32^\circ + \angle BDC$$

$$\angle BDC = 86^\circ$$

FINAL ANSWER

86°.

PAST PAPER QUESTION Q#: 14 Marks: 4

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

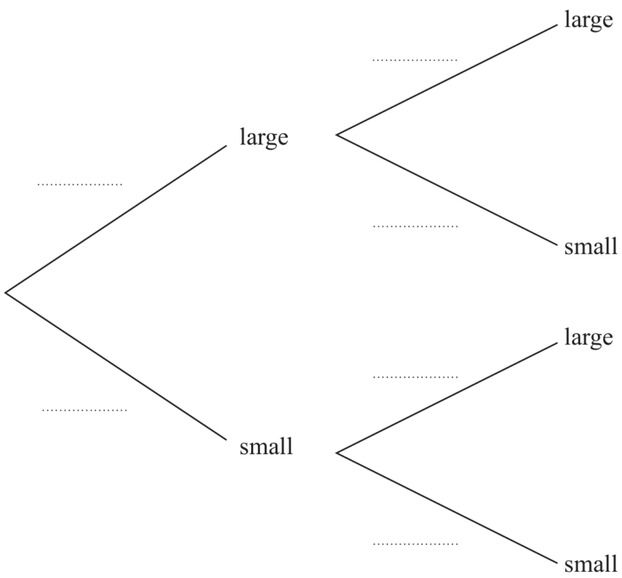
Jan 2020 P2H - Jan 2020 | P2H

14 There are 20 glasses in a cupboard.

13 of the glasses are large
7 of the glasses are small

Roberto takes at random two glasses from the cupboard.

(a) Complete the probability tree diagram.



(2)

(b) Work out the probability that Roberto takes two small glasses.

....

(2)

(Total for Question 14 is 4 marks)

PAST PAPER SOLUTION

Q#: 14 Marks: 4

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Probability Diagrams - Venn & Tree Diagrams. The tag is correct.

Solution

There are 13 large glasses and 7 small glasses.

$$P(\text{two small}) = \frac{7}{20} \times \frac{6}{19}$$

$$= \frac{21}{190}$$

FINAL ANSWER

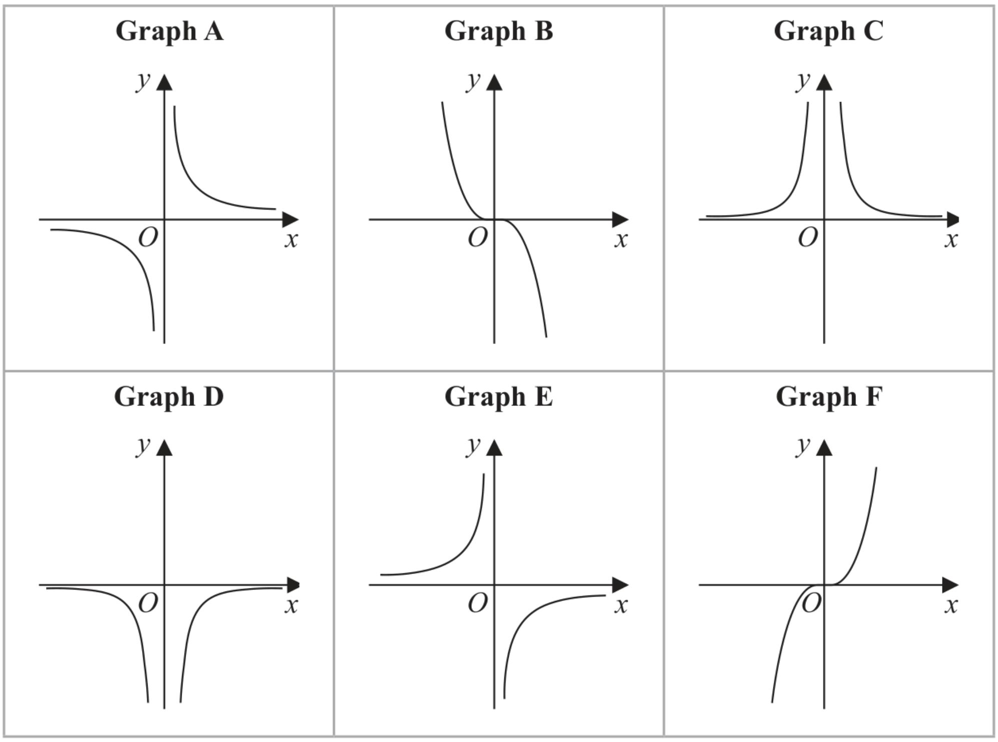
$\frac{21}{190}$.

PAST PAPER QUESTION **Q#: 15** **Marks: 3**

TOPIC: **Graphs of Functions**

Jan 2020 P2H - Jan 2020 | P2H

15 Here are six graphs.



Complete the table below with the letter of the graph that could represent each given equation.

Write your answers on the dotted lines.

Equation	Graph
$y = \dfrac{2}{x^2}$	....
$y = -\dfrac{1}{2}x^3$	....
$y = -\dfrac{5}{x}$	....

(Total for Question 15 is 3 marks)

PAST PAPER SOLUTION

Q#: 15 Marks: 3

TOPIC: **Graphs of Functions**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION

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Topic check. Correct. This is matching equations to graph shapes.

Solution

For

$$y = -\frac{2}{x^2}$$

the graph is negative on both sides of the y-axis and is symmetrical in the y-axis.
This is Graph D.
For

$$y = -\frac{1}{2}x^3$$

the graph is a decreasing cubic through the origin.
This is Graph B.
For

$$y = \frac{5}{x}$$

the graph is a positive reciprocal graph in quadrants 1 and 3.
This is Graph A.

FINAL ANSWER
D, B, A.

PAST PAPER QUESTION

Q#: 16 Marks: 4

TOPIC: **Rearranging Formulas**

Jan 2020 P2H - Jan 2020 | P2H

16 Make x the subject of $y = \sqrt{\dfrac{x+1}{x-4}}$

....

(Total for Question 16 is 4 marks)

PAST PAPER SOLUTION

Q#: 16 Marks: 4

TOPIC: **Rearranging Formulas**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION

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Topic check. Rearranging formulas. The tag is correct.

Solution

$$y = \sqrt{\frac{x+1}{x-4}}$$

Square both sides:

$$y^2 = \frac{x+1}{x-4}$$

$$y^2(x-4) = x+1$$

$$y^2x - 4y^2 = x+1$$

$$x(y^2-1) = 4y^2+1$$

FINAL ANSWER

$x = \frac{4y^2+1}{y^2-1}$

PAST PAPER QUESTION

Topic: **Algebraic Proof**

Jan 2020 P2H - Jan 2020 | P2H

Q#: 17 Marks: 3

17 Prove that the difference between two consecutive square numbers is always an odd number.
Show clear algebraic working.

(Total for Question 17 is 3 marks)

PAST PAPER SOLUTION

Q#: 17 Marks: 3

TOPIC: **Algebraic Proof**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Algebraic proof. The tag is correct.

Solution

Let the two consecutive square numbers be n^2 and $(n+1)^2$.

$$(n+1)^2 - n^2 = n^2 + 2n + 1 - n^2 = 2n + 1$$

$2n+1$ is one more than an even number, so it is always odd.

FINAL ANSWER

The difference is always odd.

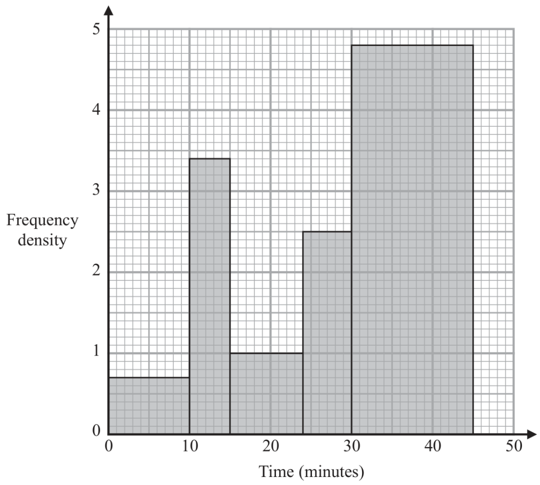
PAST PAPER QUESTION

Q#: 18 Marks: 5

TOPIC: **Histograms**

Jan 2020 P2H - Jan 2020 | P2H

18 The histogram gives information about the times, in minutes, that some customers spent in a supermarket.



(a) Work out an estimate for the proportion of these customers who spent between 17 minutes and 35 minutes in the supermarket.

....

(3)

One of the customers is selected at random.

Given that this customer had spent more than 30 minutes in the supermarket,

(b) find the probability that this customer spent more than 36 minutes in the supermarket.

....

(2)

(Total for Question 18 is 5 marks)

PAST PAPER SOLUTION

Q#: 18 Marks: 5

TOPIC: **Histograms**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION

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Topic check. Histograms. The tag is correct.

Solution

Total frequency:

$$10(0.7) + 5(3.4) + 9(1) + 6(2.5) + 15(4.8) = 120$$

Between 17 and 35 minutes:

$$7(1) + 6(2.5) + 5(4.8) = 46$$

$$\text{proportion} = \frac{46}{120} = \frac{23}{60}$$

Given the customer spent more than 30 minutes, they are in the 30 to 45 minute class. More than 36 minutes is 9 minutes out of the 15-minute class:

$$\frac{9}{15} = \frac{3}{5}$$

FINAL ANSWER

$\frac{23}{60}$, then $\frac{3}{5}$.

PAST PAPER QUESTION

TOPIC: **Coordinate Geometry**

Jan 2020 P2H - Jan 2020 | P2H

Q#: 19 **Marks: 5**

19 (a) Write down an equation of a line that is parallel to the line with equation $y = 7 - 4x$

....

(1)

The line **L** passes through the points with coordinates (–3, 1) and (2, –2)

(b) Find an equation of the line that is perpendicular to **L** and passes through the point with coordinates (–6, 4)
Give your answer in the form $ax + by + c = 0$ where a, b and c are integers.

....

(4)

(Total for Question 19 is 5 marks)

PAST PAPER SOLUTION

Q#: 19 Marks: 5

TOPIC: **Coordinate Geometry**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION

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Topic check. Coordinate geometry. The tag is correct.

Solution

(a)

$$y = 7 - 4x$$

A parallel line has the same gradient, -4. One possible answer is

$$y = -4x$$

(b) The gradient of line L through $(-3, 1)$ and $(2, -2)$ is

$$\frac{-2 - 1}{2 - (-3)} = -\frac{3}{5}$$

A perpendicular line has gradient $\frac{5}{3}$.
Through $(-6, 4)$:

$$y - 4 = \frac{5}{3}(x + 6)$$

Multiply by 3:

$$3y - 12 = 5x + 30$$

$$5x - 3y + 42 = 0$$

FINAL ANSWER

For example $y = -4x$, and $5x - 3y + 42 = 0$.

PAST PAPER QUESTION

TOPIC: **Surds**

Jan 2020 P2H - Jan 2020 | P2H

Q#: 20 Marks: 3

20 The area of a rectangle is 18 cm^2

The length of the rectangle is $\left(\sqrt{7}+1\right)$ cm.

Without using a calculator and showing each stage of your working,

find the width of the rectangle.
Give your answer in the form $a\sqrt{b}+c$ where a, b and c are integers.

.... cm

(Total for Question 20 is 3 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 3

TOPIC: **Surds**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Surds. The tag is correct.

Method

$$\text{width} = \frac{18}{\sqrt{7}+1}$$

Rationalise the denominator:

$$\frac{18}{\sqrt{7}+1} \times \frac{\sqrt{7}-1}{\sqrt{7}-1} = \frac{18(\sqrt{7}-1)}{7-1}$$

$$= \frac{18(\sqrt{7}-1)}{6} = 3(\sqrt{7}-1)$$

$$= 3\sqrt{7} - 3$$

FINAL ANSWER

$3\sqrt{7} - 3$ cm

PAST PAPER QUESTION

TOPIC: **Surds**

Jan 2020 P2H - Jan 2020 | P2H

Q#: 20 Marks: 3

20 The area of a rectangle is 18 cm^2

The length of the rectangle is $\left(\sqrt{7} + 1\right)$ cm.

Without using a calculator and showing each stage of your working,

find the width of the rectangle.
Give your answer in the form $a\sqrt{b} + c$ where a, b and c are integers.

.... cm

(Total for Question 20 is 3 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 3

TOPIC: **Surds**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Right-Angled Triangles to Surds.

Method

Area of rectangle:

$$18 \text{ cm}^2$$

Length:

$$\sqrt{7} + 1$$

Width:

$$\frac{18}{\sqrt{7}+1}$$

Rationalise the denominator:

$$\frac{18}{\sqrt{7}+1} \times \frac{\sqrt{7}-1}{\sqrt{7}-1}$$

$$= \frac{18(\sqrt{7}-1)}{7-1}$$

$$= \frac{18(\sqrt{7}-1)}{6}$$

$$= 3(\sqrt{7}-1)$$

$$= 3\sqrt{7} - 3$$

FINAL ANSWER

$\boxed{3\sqrt{7} - 3 \text{ cm}}$

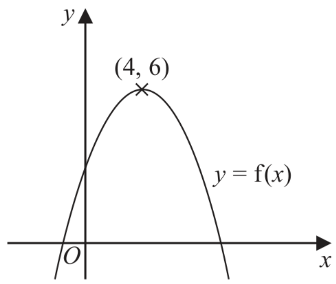
PAST PAPER QUESTION

Q#: 21 Marks: 4

TOPIC: **Transformations of Graphs**

Jan 2020 P2H - Jan 2020 | P2H

21 The diagram shows a sketch of part of the curve with equation $y = f(x)$



There is one maximum point on this curve.
The coordinates of this maximum point are (4, 6)

(a) Write down the coordinates of the maximum point on the curve with equation

(i) $y = f(x + 4)$

(.............................. , ...)

(ii) $y = f(2x)$

(.............................. , ...)

(2)

The equation of a curve **C** is $y = x^2 + 3x + 4$

The curve **C** is transformed to curve **S** under the translation $\begin{pmatrix} 4 \\ 6 \end{pmatrix}$

(b) Find an equation of curve **S**.

You do not need to simplify the equation.

....

(2)

(Total for Question 21 is 4 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 4

TOPIC: **Transformations of Graphs**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The maximum point of $y = f(x)$ is $(4, 6)$.
For

$$y = f(x + 4)$$

the graph moves 4 units left:

$$(4, 6) \to (0, 6)$$

For

$$y = f(2x)$$

the x-coordinate is divided by 2:

$$(4, 6) \to (2, 6)$$

For part (b),

$$y = x^2 + 3x + 4$$

The translation vector is

$$\begin{pmatrix} 4 \\ 6 \end{pmatrix}$$

So the graph moves 4 units right and 6 units up.
Replace x by $x - 4$, then add 6:

$$y = (x - 4)^2 + 3(x - 4) + 4 + 6$$

$$y = (x - 4)^2 + 3(x - 4) + 10$$

FINAL ANSWER
$(0, 6)$, $(2, 6)$, and $y = (x - 4)^2 + 3(x - 4) + 10$

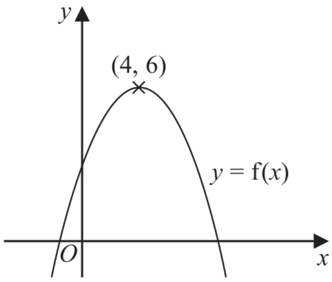
PAST PAPER QUESTION

Q#: 21 Marks: 4

TOPIC: **Transformations of Graphs**

Jan 2020 P2H - Jan 2020 | P2H

21 The diagram shows a sketch of part of the curve with equation $y = f(x)$



There is one maximum point on this curve.
The coordinates of this maximum point are (4, 6)

(a) Write down the coordinates of the maximum point on the curve with equation

(i) $y = f(x + 4)$

(.............................. , ...)

(ii) $y = f(2x)$

(.............................. , ...)

(2)

The equation of a curve **C** is $y = x^2 + 3x + 4$

The curve **C** is transformed to curve **S** under the translation $\begin{pmatrix} 4 \\ 6 \end{pmatrix}$

(b) Find an equation of curve **S**.

You do not need to simplify the equation.

....

(2)

(Total for Question 21 is 4 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 4

TOPIC: **Transformations of Graphs**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The maximum point of $y = f(x)$ is $(4, 6)$.
For

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For

$$y = f(2x)$$

the x-coordinate is divided by 2:

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The translation vector is

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So the graph moves 4 units right and 6 units up.
Replace x by $x - 4$, then add 6:

$$y = (x - 4)^2 + 3(x - 4) + 4 + 6$$

$$y = (x - 4)^2 + 3(x - 4) + 10$$

FINAL ANSWER

$(0, 6)$, $(2, 6)$, and $y = (x - 4)^2 + 3(x - 4) + 10$

PAST PAPER QUESTION

Q#: 22 **Marks: 5**

TOPIC: **Coordinate Geometry**

Jan 2020 P2H - Jan 2020 | P2H

22 The line with equation $y = x + 2$ intersects the curve with equation $x^2 + y^2 - 2y = 24$ at the points A and B.

Find the coordinates of A and B.
Show clear algebraic working.

(.............................. , ...)

(.............................. , ...)

(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 5

TOPIC: **Coordinate Geometry**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The line is

$$y = x + 2$$

Substitute this into the circle equation:

$$x^2 + y^2 - 2y = 24$$

$$x^2 + (x+2)^2 - 2(x+2) = 24$$

$$x^2 + x^2 + 4x + 4 - 2x - 4 = 24$$

$$2x^2 + 2x - 24 = 0$$

$$x^2 + x - 12 = 0$$

$$(x+4)(x-3) = 0$$

So

$$x = -4 \quad \text{or} \quad x = 3$$

Using $y = x + 2$:

$$x = -4 \Rightarrow y = -2$$

$$x = 3 \Rightarrow y = 5$$

FINAL ANSWER

$(-4, -2)$ and $(3, 5)$

PAST PAPER QUESTION

Q#: 22 Marks: 5

TOPIC: **Coordinate Geometry**

Jan 2020 P2H - Jan 2020 | P2H

22 The line with equation $y = x + 2$ intersects the curve with equation $x^2 + y^2 - 2y = 24$ at the points A and B.

Find the coordinates of A and B.
Show clear algebraic working.

(.............................. , ...)

(.............................. , ...)

(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 5

TOPIC: **Coordinate Geometry**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The line is

$$y = x + 2$$

Substitute this into the circle equation:

$$x^2 + y^2 - 2y = 24$$

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$$x^2 + x^2 + 4x + 4 - 2x - 4 = 24$$

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So

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Using $y = x + 2$:

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FINAL ANSWER

$(-4, -2)$ and $(3, 5)$

PAST PAPER QUESTION

Q#: 23 Marks: 3

TOPIC: **Vectors**

Jan 2020 P2H - Jan 2020 | P2H

23

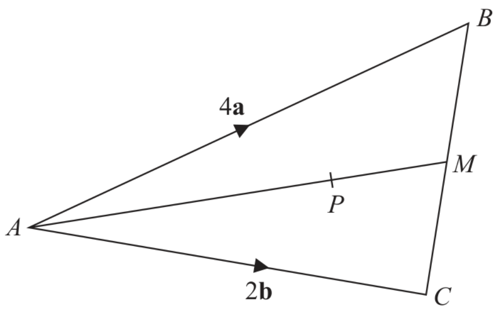

Diagram **NOT** accurately drawn

ABC is a triangle.
The midpoint of BC is M.
P is a point on AM.

$\overrightarrow{AB} = 4\mathbf{a}$

$\overrightarrow{AC} = 2\mathbf{b}$

$\overrightarrow{AP} = \frac{3}{2}\mathbf{a} + \frac{3}{4}\mathbf{b}$

Find the ratio $AP : PM$

....

(Total for Question 23 is 3 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 3

TOPIC: **Vectors**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Ratio Toolkit to Vectors.

Method

Since M is the midpoint of BC,

$$\overrightarrow{AM} = \frac{\overrightarrow{AB} + \overrightarrow{AC}}{2}$$

$$\overrightarrow{AM} = \frac{4a + 2b}{2} = 2a + b$$

Given

$$\overrightarrow{AP} = \frac{3}{2}a + \frac{3}{4}b$$

Factorise:

$$\overrightarrow{AP} = \frac{3}{4}(2a + b) = \frac{3}{4}\overrightarrow{AM}$$

So AP is $\frac{3}{4}$ of AM, leaving PM as $\frac{1}{4}$ of AM.

FINAL ANSWER

$AP : PM = 3 : 1$

PAST PAPER QUESTION Q#: 23 Marks: 3

TOPIC: **Vectors**

Jan 2020 P2H - Jan 2020 | P2H

23

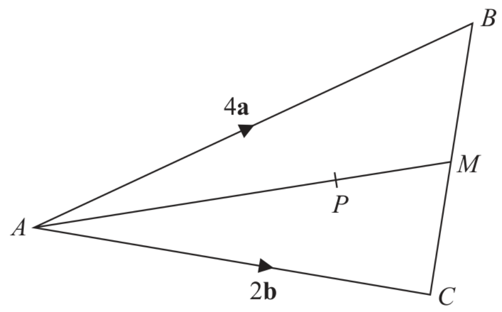

Diagram **NOT** accurately drawn

ABC is a triangle.
The midpoint of BC is M.
P is a point on AM.

$\overrightarrow{AB} = 4\mathbf{a}$

$\overrightarrow{AC} = 2\mathbf{b}$

$\overrightarrow{AP} = \frac{3}{2}\mathbf{a} + \frac{3}{4}\mathbf{b}$

Find the ratio $AP : PM$

....

(Total for Question 23 is 3 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 3

TOPIC: **Vectors**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Ratio Toolkit to Vectors.

Method

Since M is the midpoint of BC,

$$\overrightarrow{AM} = \frac{\overrightarrow{AB} + \overrightarrow{AC}}{2}$$

$$\overrightarrow{AM} = \frac{4a + 2b}{2} = 2a + b$$

Given

$$\overrightarrow{AP} = \frac{3}{2}a + \frac{3}{4}b$$

Factorise:

$$\overrightarrow{AP} = \frac{3}{4}(2a + b) = \frac{3}{4}\overrightarrow{AM}$$

So AP is $\frac{3}{4}$ of AM, leaving PM as $\frac{1}{4}$ of AM.

FINAL ANSWER

$AP : PM = 3 : 1$

PAST PAPER QUESTION

Q#: 24 Marks: 4

TOPIC: **Algebraic Fractions**

Jan 2020 P2H - Jan 2020 | P2H

24 Express

$$\left(\frac{4}{2x-5}-\frac{3}{2x-3}\right)\div\frac{9x-4x^3}{6x^2-17x+5}$$

as a single fraction in its simplest form.

....

(Total for Question 24 is 4 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 4

TOPIC: **Algebraic Fractions**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Sine/Cosine Rule to Algebraic Fractions.

Method

$$\left(\frac{4}{2x-5}-\frac{3}{2x-3}\right)\div\frac{9x-4x^3}{6x^2-17x+5}$$

First simplify the bracket:

$$\frac{4}{2x-5}-\frac{3}{2x-3}=\frac{4(2x-3)-3(2x-5)}{(2x-5)(2x-3)}$$
$$=\frac{8x-12-6x+15}{(2x-5)(2x-3)}$$
$$=\frac{2x+3}{(2x-5)(2x-3)}$$

Now factor the second fraction:

$$9x-4x^3=-x(2x-3)(2x+3)$$

$$6x^2-17x+5=(2x-5)(3x-1)$$

Dividing by a fraction means multiplying by its reciprocal:

$$\frac{2x+3}{(2x-5)(2x-3)}\times\frac{(2x-5)(3x-1)}{-x(2x-3)(2x+3)}$$

Cancel common factors:

$$=-\frac{3x-1}{x(2x-3)^2}$$
$$=\frac{1-3x}{x(2x-3)^2}$$

FINAL ANSWER

$$\boxed{\frac{1-3x}{x(2x-3)^2}}$$

PAST PAPER QUESTION

Q#: 24 **Marks: 4**

TOPIC: **Algebraic Fractions**

Jan 2020 P2H - Jan 2020 | P2H

24 Express

$$\left(\frac{4}{2x-5}-\frac{3}{2x-3}\right)\div\frac{9x-4x^3}{6x^2-17x+5}$$

as a single fraction in its simplest form.

....

(Total for Question 24 is 4 marks)

PAST PAPER SOLUTION

Q#: 24 **Marks: 4**

TOPIC: **Algebraic Fractions**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Sine/Cosine Rule to Algebraic Fractions.

Method

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First simplify the bracket:

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Cancel common factors:

$$\begin{aligned}&=-\frac{3x-1}{x(2x-3)^2}\\&=\frac{1-3x}{x(2x-3)^2}\end{aligned}$$

FINAL ANSWER

$$\boxed{\frac{1-3x}{x(2x-3)^2}}$$

PAST PAPER QUESTION

Q#: 25 Marks: 3

TOPIC: **Sequences**

Jan 2020 P2H - Jan 2020 | P2H

25 Mario is going to save \$50 in the year 2021

He is going to continue to save, up to and including the year 2070, by increasing the amount he saves each year by \$$k$

Mario will save a total of \$33 125 from 2021 to 2070

Work out the value of k.

k = ...

(Total for Question 25 is 3 marks)

PAST PAPER SOLUTION

Q#: 25 Marks: 3

TOPIC: **Sequences**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Right-Angled Triangles to Sequences.

Method

Mario saves from 2021 up to and including 2070.
Number of years:

$$2070 - 2021 + 1 = 50$$

This is an arithmetic sequence.
First term:

$$50$$

Common difference:

$$k$$

Last term:

$$50 + 49k$$

Use the arithmetic series formula:

$$S_n = \frac{n}{2}(a + l)$$

$$33125 = \frac{50}{2}(50 + (50 + 49k))$$

$$33125 = 25(100 + 49k)$$

$$1325 = 100 + 49k$$

$$49k = 1225$$

$$k = 25$$

FINAL ANSWER

$\boxed{k = 25}$

PAST PAPER QUESTION **Q#: 25** **Marks: 3**

TOPIC: **Sequences**

Jan 2020 P2H - Jan 2020 | P2H

25 Mario is going to save \$50 in the year 2021

He is going to continue to save, up to and including the year 2070, by increasing the amount he saves each year by \$$k$

Mario will save a total of \$33 125 from 2021 to 2070

Work out the value of k.

k = ...

(Total for Question 25 is 3 marks)

PAST PAPER SOLUTION

Q#: 25 Marks: 3

TOPIC: **Sequences**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Right-Angled Triangles to Sequences.

Method

Mario saves from 2021 up to and including 2070.
Number of years:

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This is an arithmetic sequence.
First term:

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Common difference:

$$k$$

Last term:

$$50 + 49k$$

Use the arithmetic series formula:

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$$33125 = \frac{50}{2}\left(50 + (50 + 49k)\right)$$

$$33125 = 25(100 + 49k)$$

$$1325 = 100 + 49k$$

$$49k = 1225$$

$$k = 25$$

FINAL ANSWER

$\boxed{k = 25}$

PAST PAPER QUESTION

Q#: 26 Marks: 6

Topic: **Volume & Surface Area**

Jan 2020 P2H - Jan 2020 | P2H

26 Here is a sector, *AOB*, of a circle with centre *O* and angle $AOB = x°$

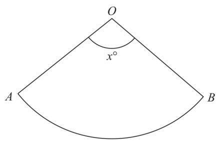


Diagram **NOT** accurately drawn

The sector can form the curved surface of a cone by joining *OA* to *OB*.

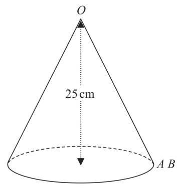


Diagram **NOT** accurately drawn

The height of the cone is 25 cm.
The volume of the cone is 1600 cm^3

Work out the value of *x*.
Give your answer correct to the nearest whole number.

$x =$ ....

(Total for Question 26 is 6 marks)

PAST PAPER SOLUTION Q#: 26 Marks: 6

TOPIC: **Volume & Surface Area**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Moved to Volume & Surface Area.

Method

Let the radius of the base of the cone be r, and let the slant height be l.
The volume of the cone is

$$\frac{1}{3}\pi r^2 h = 1600$$

The height is 25 cm, so

$$\frac{1}{3}\pi r^2 (25) = 1600$$

$$r^2 = \frac{4800}{25\pi}$$

$$r = 7.8176\ldots$$

The slant height is found using Pythagoras:

$$l^2 = r^2 + 25^2$$

$$l = 26.1938\ldots$$

When the sector is joined to form the cone, the arc length of the sector becomes the circumference of the cone base.

$$\frac{x}{360} \times 2\pi l = 2\pi r$$

Cancel 2π:

$$\frac{x}{360} l = r$$

$$x = \frac{360r}{l}$$

$$x = \frac{360(7.8176\ldots)}{26.1938\ldots} = 107.4433\ldots$$

FINAL ANSWER

$\boxed{107^\circ}$

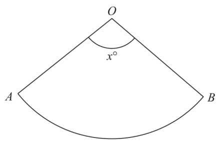
PAST PAPER QUESTION

Q#: 26 Marks: 6

TOPIC: **Volume & Surface Area**

Jan 2020 P2H - Jan 2020 | P2H

26 Here is a sector, *AOB*, of a circle with centre *O* and angle *AOB* = $x°$



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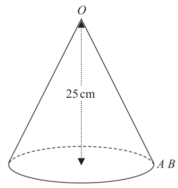

Diagram **NOT** accurately drawn

The height of the cone is 25 cm.
The volume of the cone is 1600 cm^3

Work out the value of x.
Give your answer correct to the nearest whole number.

x = ...

(Total for Question 26 is 6 marks)

PAST PAPER SOLUTION

Q#: 26 Marks: 6

TOPIC: **Volume & Surface Area**

Jan 2020 P2H - Jan 2020 | P2H

WORKED SOLUTION

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FINAL ANSWER

$\boxed{107^\circ}$